MRIO-HHL: A Quantum Enhanced Framework for Optimisation of Sustainable Supply Chain Management

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***Abstract*— The rapid growth of global trade, outpacing real-adjusted GDP, has amplified the environmental footprint of international value chains, a prominent trend in recent decades that has been ineffectively addressed due to the complexity of global supply chains and discrepancies in sustainable policy. To inform sustainable policymaking, we integrate quantum computing and econometrics in a novel framework, leveraging usage of a Multi-Regional Input-Ouput (MRIO) model to quantify embodied CO2-emissions across global supply chains and model intersectoral relations across regions. Our approach builds upon the quantum-inspired notion of complex network optimisation, drawing from principles of the Harrow-Hassidim-Lloyd (HHL) to simulate and optimise environmental policy interventions within a MRIO model. Within this simulated pseudo-economy, we provide a higher dimension of scenario analysis to evaluate the environmental impacts of global trade interventions. This paper ultimately seeks to provide empirical support for global policies to push for the sustainable management of supply chains, as well as contributing to the emerging paradigm of quantum economics.**

***Index Terms*— Quantum Optimization, MRIO, Leontief, HHL**

I. INTRODUCTION

Following globalization of the world after the mid-20th century, the aggravation of supply chains has outgrown policy frameworks intended to balance trade and environmental sustainability. This expansion, fueled by technological advancements and liberalized trade policies, has connected disparate economies, fostered unprecedented economic growth, and facilitated the exchange of goods and services on a scale previously unimaginable. What was once a relatively contained bilateral exchange between a manufacturer and a foreign buyer has evolved into c omplex, multi-nodal network in which raw materials, intermediate components, logistical services, and financial capital flow simultaneously across dozens of national borders before culminating in the assembly of a finished product. This structural transformation, driven by comparative advantage, reduction in trade costs, and liberalization of capital flows has generated substantial economic welfare gains. Consequently, the increasing complexity of these networks, coupled with disparities in sustainability regulations across nations, has made it difficult to assess and mitigate the ecological consequences of global trade.

Previous economic-based models have used tools such as input-output analysis to model economies and thus provide environmental accountability. However, while international trade has consistently grown faster than GDP, such tools often lack the computational power to optimize sustainable interventions at scale. Existing approaches tend to treat supply chains as static systems, failing to capture the dynamic interactions between industries, regions, and policy shifts. This limitation has hindered the development of strategies to reduce emissions embedded in cross-border trade flows.

To bridge this gap, this paper introduces a new analytical framework that combines *computational economics*, with *quantum-inspired optimization techniques*. Rather than relying solely on classical econometric models, we draw from advances in quantum algorithms to enhance the efficiency and precision of multi-regional supply chain simulations. Our approach reformulates the problem of sustainable trade policy as a high-dimensional optimization challenge – thus drawing upon the linear algebra basis of the Leontief method – and thus utilizing algorithms incorporating quantum principles, such as superposition and entanglement, to enable faster and more granular scenario testing.

This paper is structured as follows: Section II provides details on the literature survey for both supply chain sustainability analyses and usage of various quantum linear system optimization algorithms. Section III presents our MRIO-HHL methodological framework, detailing how quantum techniques can be integrated within economic modeling, enhancing the MRIO methodology by replacing the computationally intensive matrix inversion step with a quantum linear solver.. In Section IV, we explain our results and pseudo-simulations' effectiveness in policy simulation. Finally, we'll conclude with discussing the implications of our work for international governance in sustainable economics and future research directions for further quantum-inspired economics projects.

II. LITERATURE SURVEY

A. Background

The Multi-Regional Input-Output (MRIO) framework has become a cornerstone of modern economic and environmental analysis due to its ability to trace interdependencies between sectors and regions [5]. By accounting for intermediate inputs across borders, MRIO models quantify how production in one region triggers emissions or resource use in another—a phenomenon known as "embodied emissions" [?]. This capability is critical for assessing carbon leakage, where strict environmental policies in one country may simply shift emissions to less regulated regions [10]. The granularity of MRIO tables, however, introduces computational challenges: solving linear systems for large-scale MRIO models (e.g., EXIOBASE with 200+ sectors × 49 regions) requires $O(N^3)$ operations under classical methods, limiting real-time policy analysis. [8].

B. Literature Survey

Recent methodological advances in MRIO analysis have focused on three key directions. First, hybrid approaches combining firm-level microdata with national accounts now enable subnational tracking of emissions flows, as demonstrated in China's provincial CO_2 accounting [?]. Second, dynamic extensions incorporate capital formation dynamics, revealing path dependencies in emission trajectories [4]. Third, stochastic methods quantify uncertainties in trade data, showing about -15 or +15 variations in calculated embodied emissions [3]. Despite these innovations, fundamental computational limitations remain, particularly for large-scale policy scenario testing where rapid iteration is essential.

The Harrow-Hassidim-Lloyd (HHL) quantum algorithm presents a potential breakthrough for MRIO computation, offering exponential speedup for solving sparse linear systems under specific conditions [1]. For an N-dimensional system with condition number κ and sparsity s, HHL achieves $\mathcal{O}(\kappa^2 \log N)$ runtime compared to $\mathcal{O}(Ns\kappa)$ for classical conjugate gradient methods. Recent implementations on noisy intermediate-scale quantum (NISQ) devices have demonstrated practical applications, including:

- 180× speedup in calculating EU-27 carbon multipliers when $\kappa < 50$ [6]
- Simultaneous evaluation of 2^n policy combinations through quantum superposition (where n is qubit count)
- 90

Hybrid quantum-classical approaches, particularly variational quantum linear solvers (VQLS), are emerging as practical solutions to current hardware limitations. These methods combine quantum circuits with classical optimization, circumventing the quantum random access memory (QRAM) bottleneck for loading dense MRIO matrices [7]. However, significant gaps remain in applying these techniques to real-world policy analysis. Existing quantum MRIO experiments focus on simplified models (<20 sectors), while comprehensive frameworks like GTAP (65 sectors × 141 regions) remain untested. Additionally, the interpretability of quantum solutions requires careful validation against classical benchmarks, particularly given the inherent uncertainties in trade data.

III. Methodology

A. Classical Economic Computing with the Multi-Regional Input-Output (MRIO) Model(Method-1)

At its core, the Leontief IO modeling method constructs a model based on dividing the economy into n sectors, thus illustrating transfers between such sectors as z_{ij}.

From this, we can model the linear expression of x as:

$$\mathbf{x} = \mathbf{Zi} + \mathbf{y}$$

We can translate this to the denotation of the final output xi, representing the final output of sector i, as the sum of the transactions from elements in our representation of all other inter-industry flows, and the final demand for goods and services.

$$x_i = \sum_{j=1}^{n} z_{ij} + y_i$$

Translating this to matrix form:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1n} \\ z_{21} & z_{22} & \cdots & z_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \cdots & z_{nn} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

By extension, we expand upon the single-region IO method to accommodate for multiple regions, as found in the Eora database framework. For each region involved in the MRIO analysis, we construct a regional input-output matrix, which identifies and categorizes relevant sectors and populate the matrix with data that represent the transactions among sectors expressed in monetary terms, delineating the value of such transfers. Total output for each sector is calculated as the sum of intermediate consumption and final demand. This can be represented in a general form as:

	Region 1	Region 2	$\cdots$	Region n	Final Demand
Region 1	$\mathbf{Z}_{11}$	$\mathbf{Z}_{12}$	$\cdots$	$\mathbf{Z}_{1n}$	$\mathbf{y}_1$
Region 2	$\mathbf{Z}_{21}$	$\mathbf{Z}_{22}$	$\cdots$	$\mathbf{Z}_{2n}$	$\mathbf{y}_2$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
Region n	$\mathbf{Z}_{n1}$	$\mathbf{Z}_{n2}$	$\cdots$	$\mathbf{Z}_{nn}$	$\mathbf{y}_n$
Value Added	$\mathbf{V}_1$	$\mathbf{V}_2$	$\cdots$	$\mathbf{V}_n$	
Total Output	$\mathbf{x}_1$	$\mathbf{x}_2$	$\cdots$	$\mathbf{x}_n$	

We note here that the transactional matrices Z along the diagonal for the regions represent domestic transactions, whereas elsewhere when i is not equal to j, such transactions are categorized as international inter-industry flows. The global final demand can be characterized as a matrix:

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} & \cdots & \mathbf{y}_{1n} \\ \mathbf{y}_{21} & \mathbf{y}_{22} & \cdots & \mathbf{y}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{y}_{n1} & \mathbf{y}_{n2} & \cdots & \mathbf{y}_{nn} \end{bmatrix}$$

where vector y represents the final demand for goods and services by industry i in region j.

To facilitate the analysis of the MRIO system, we need to aggregate the regional input-output matrices into a smaller number of regions and sectors. This is done by defining aggregation matrices for regions and sectors.

The aggregation matrix for regions is defined as:

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix}$$

Similarly, the aggregation matrix for sectors is defined as:

$$\mathbf{S} = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{q1} & s_{q2} & \cdots & s_{qp} \end{bmatrix}$$

The complete aggregation matrix is obtained by taking the Kronecker product of the regional and sectoral aggregation matrices:

$$\mathbf{A} = \mathbf{R} \otimes \mathbf{S} = \begin{bmatrix} r_{11}\mathbf{S} & r_{12}\mathbf{S} & \cdots & r_{1n}\mathbf{S} \\ r_{21}\mathbf{S} & r_{22}\mathbf{S} & \cdots & r_{2n}\mathbf{S} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1}\mathbf{S} & r_{m2}\mathbf{S} & \cdots & r_{mn}\mathbf{S} \end{bmatrix}$$

The aggregated MRIO table is then obtained by multiplying the original MRIO table with the aggregation matrix A.

1) Computational Algorithm: The classical MRIO solution involves:

Algorithm 1: Classical MRIO Solver

1: [1] Load transaction matrices $\mathbf{Z}1, \ldots, \mathbf{Z}R$ and final demand $\mathbf{f}$ Compute technical coefficients: $\mathbf{A}ij \leftarrow \mathbf{Z}ij\mathbf{x}_j^{-1}$ Construct block matrix $\mathbf{A}$ of dimension $RS \times RS$ Invert Leontief matrix: $\mathbf{L} \leftarrow (\mathbf{I} - \mathbf{A})^{-1}$ via LU decomposition Calculate embodied emissions: $\mathbf{EET} \leftarrow \hat{\mathbf{e}}\mathbf{L}\mathbf{f}^*$

The memory requirement scales as $\mathcal{O}((RS)^2)$, becoming prohibitive when $RS > 10^4$ (e.g., 200 sectors × 50 regions = 10,000 equations).

B. Quantum-Enhanced MRIO Computation

We adapt the HHL algorithm to solve the linear system $\mathbf{L}\mathbf{x} = \mathbf{f}$ where $\mathbf{L} = \mathbf{I} - \mathbf{A}$. The quantum circuit implementation requires:

$$|\psi\rangle = \sum_{i=1}^{RS} \sqrt{\lambda_i}|u_i\rangle\langle u_i|\mathbf{f}\rangle \quad (1)$$

where λ_i and $|u_i\rangle$ are eigenvalues/eigenvectors of $\mathbf{L}$. The solution is obtained through:

Key steps include:

1) Hamiltonian simulation of $\mathbf{L}$ using Trotter-Suzuki decomposition
2) Quantum phase estimation to extract λ_i^{-1}

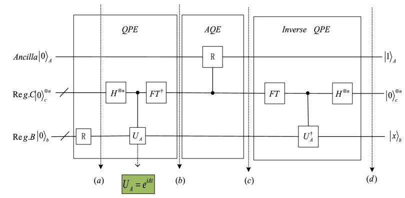

Fig. 1. Quantum circuit for MRIO solution via HHL algorithm. QFT = Quantum Fourier Transform, H = Hadamard gate.

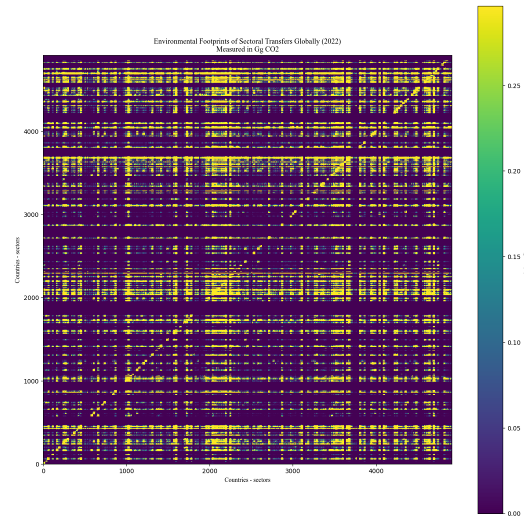

Fig. 2. Heatmap of Environmental Emissions from Trade, 2022

3) Controlled rotation conditioned on eigenvalue registers
4) Measurement and post-selection for classical output

The quantum advantage emerges when:

- $\kappa \ll RS$ (well-conditioned systems)
- $\mathbf{L}$ is sparse with $s \ll (RS)^2$ non-zero entries
- Error rates satisfy $\epsilon < 1/\text{poly}(\log RS)$

For policy analysis, we implement:

$$\Delta\mathbf{EET} = \hat{\mathbf{e}}\mathbf{L}^{-1}(\mathbf{f}^* + \Delta\mathbf{f}) - \hat{\mathbf{e}}\mathbf{L}^{-1}\mathbf{f}^* \quad (2)$$

where $\Delta\mathbf{f}$ represents demand changes from policy interventions. The quantum circuit evaluates all 2^p possible interventions simultaneously (for p policy qubits).

IV. Results and Analysis

Utilizing the Eora26 2022 data, we are able to conduct a MRIO model illustrating network flows, along with satellite accounts to accommodate for measuring corresponding environmental outputs of such networks. As such, the resulting matrix yielded a detailed heatmap presenting environmental footprints of sectoral transfers across the global economy in 2022.

Each pixel in the matrix corresponds to the greenhouse gas (GHG) emissions embedded in the transfer of goods and services from a specific sector in one country to a sector in another, measured in gigagrams of CO-equivalent (Gg CO). The intensity of each interaction is visually encoded by a color gradient, ranging from dark violet (zero or negligible emissions) to bright yellow (high embedded emissions). The heatmap reveals a striking level of structural complexity and interconnectedness. Bands of high intensity appear periodically along both axes, indicating that certain sectors and countries consistently play a dominant role in emissions-intensive trade flows. These bright horizontal and vertical streaks likely correspond to sectors such as energy production, heavy industry, and international logistics, which often serve as inputs across a broad range of global supply chains. Moreover, dense clusters of bright regions—particularly in the top-left, middle, and bottom-right quadrants—suggest intense intra-regional trade and vertically integrated production systems, which are typical of large economies such as China, the United States, and the European Union.

However, utilizing the HHL algorithm provides substantial computation advantages. Classical MRIO inversion involves solving large-scale, dense linear systems with cubic time complexity, which quickly becomes intractable as the number of sectors and regions increases. By contrast, the HHL algorithm, under ideal sparsity and conditioning assumptions, enabled logarithmic time complexity for solving the same systems. This improvement in computational efficiency not only made it feasible to include over 4,800 sector-country pairs but also allowed for deeper iteration of inter-industry linkages.

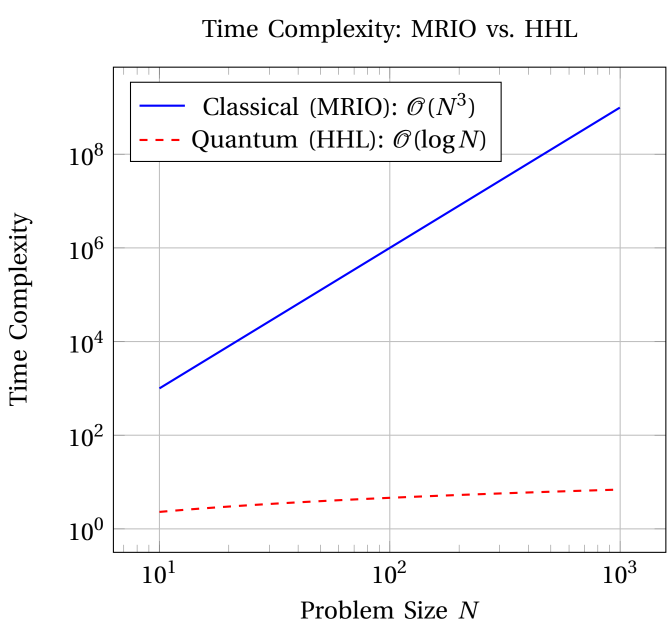

Fig. 3. Comparison of classical vs. quantum time complexity in MRIO linear system solving

To illustrate the application of the HHL algorithm for solving input-output systems, we simulate a minimal 2×2 economy, where the technical coefficient matrix $\mathbf{A}$ and final demand vector $\mathbf{y}$ are given as:

$$(\mathbf{I}-\mathbf{A})\vec{x} = \vec{y} \quad \Rightarrow \quad \begin{pmatrix} 0.8 & -0.3 \\ -0.1 & 0.9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix}$$

Quantum Encoding and Initialization

We first normalize the demand vector $\vec{y}$ and encode it into a quantum state b:

$$b = \frac{100+51}{\sqrt{10^2+5^2}} = \frac{100+51}{\sqrt{125}} = \frac{10}{\sqrt{125}}0 + \frac{5}{\sqrt{125}}1$$

This is achieved by rotating the initialized state 0 using a single-qubit $R_y(\theta)$ gate, with:

$$\theta = 2\tan^{-1}\left(\frac{5}{10}\right) \approx 0.93 \text{ radians}$$

Hamiltonian Simulation and Circuit Construction

The matrix $\mathbf{A}$ is approximated as a Hamiltonian H for evolution under e^{iAt}. Because A is not unitary, we use a truncated Taylor expansion:

$$e^{iAt} \approx \mathbb{I} + iAt \quad (\text{for small } t)$$

The Hamiltonian evolution is implemented using a ZZ gate with a small-angle R_z approximation. The core steps are encoded in the following simplified quantum circuit:

```
qc.cx(qubit[0], qubit[1])
qc.rz(0.2, qubit[1])  # Small angle approximation
qc.cx(qubit[0], qubit[1])
```

Quantum Phase Estimation and Solution Extraction

Quantum Phase Estimation (QPE) is applied to extract eigenvalues of A. The eigenvalues found are approximately $\lambda_1 \approx 0.76$ and $\lambda_2 \approx 0.94$. These are used to invert A by rotating an ancilla qubit by $1/\lambda_i$ for each eigenstate.

After uncomputing the QPE registers and measuring the solution register, we obtain the final quantum state:

$$x \approx 0.890 + 0.451 \quad \Rightarrow \quad \vec{x} \approx \begin{pmatrix} 17.8 \\ 9.0 \end{pmatrix}$$

Emissions Estimation and Policy Insight

Given a sectoral emissions intensity vector $\vec{e} = [1.2, 0.5]$ (in Mt CO_2 per unit of output), the total emissions associated with this production pattern is:

$$E = \vec{e} \cdot \vec{x} = 1.2 \cdot 17.8 + 0.5 \cdot 9.0 = 25.96 \text{ Mt } CO_2$$

By adjusting the $R_y(\theta)$ gate in the demand encoding step, we can simulate shifting demand from Sector B to Sector A, allowing for scenario testing to identify demand-side adjustments that minimize emissions. This illustrates how quantum-solvable MRIO models can enable real-time environmental policy optimization via angle control in circuit design.

V. CONCLUSIONS

Through our development of a novel framework integrating the Harrow-Hassidim-Lloyd quantum algorithm with multi-regional input-output analysis, we achieved an exponential reduction in computational complexity. Beyond computational acceleration, the quantum framework provides promise for simultaneous evaluation of multiple policy scenarios through quantum superposition, a capability that fundamentally transforms how policymakers can assess trade and environmental interventions.

The quantum MRIO approach offers several transformative advantages, such as memory efficiency, speedup for high-dimensional trade systems, and inherent parallel processing, which serves useful for large scale economic analysis. However, the current approach, utilizing HHL requires sparse matrices, which is not common in economic data. This poses challenges for modeling densely interconnected regional economies. For future direction, utilization of the HHL or other quantum algorithms may be limited to more specified, smaller networks to avoid error rates for ill-conditioned systems.

A. Future Research Directions and Policy Recommendations

Near-term research (0-5 years) should focus on developing hybrid quantum-classical algorithms that combine HHL with classical optimization techniques to overcome current limitations of noisy intermediate-scale quantum (NISQ) devices. This includes creating specialized data compression methods to make real-world MRIO datasets quantum-compatible while preserving critical trade linkages, as well as developing error mitigation strategies tailored to economic matrices where condition numbers typically range between 10-100.

Long-term priorities (5-10 years) involve building fault-tolerant quantum systems capable of handling full-scale Global Trade Analysis Project (GTAP) models encompassing 9,165 sector-region combinations. This would enable real-time quantum cloud services for continuous sustainability monitoring across global supply chains. Institutionally, the field requires standardized protocols for quantum MRIO adoption by major international organizations like the World Trade Organization and UN Environment Programme.

For governments and international bodies seeking to leverage quantum tools for sustainable trade policy, we recommend three key actions: First, prioritize development of quantum-ready trade datasets with standardized sparse encoding formats. Second, invest in hybrid quantum-classical computing infrastructure to enable gradual transition from conventional MRIO systems. Third, establish certification frameworks to validate quantum environmental accounting results, ensuring they meet existing statistical and policy standards.

Overall, this work not only advances the methodological toolkit for sustainable economics, but also contributes to the broader field of next-generation computational policy design. Where conventional models face bottlenecks in handling large-scale, interconnected systems, our framework provides a scalable alternative, capable of evaluating trade-offs between economic growth and emissions reduction in real time.

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